



# The cost of computation and supercomputers

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FLOPs, GPU hours, dollars and energy

Carlos Cotrini



# The same three questions, asked for the AI

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Was kann *es* wissen? → Weekend 5: RAG

Was soll *es* tun? → Weekend 6: Agentic AI

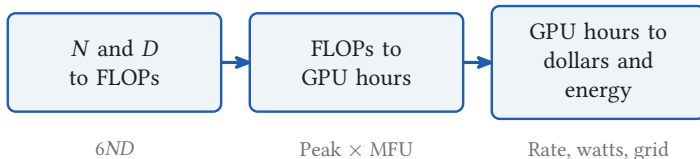
Was darf *es* hoffen? → Weekend 7: Large Scale AI what it may become, for what we spend

Weekend 5 was what it can know, Weekend 6 what it should do. And this weekend is what it may become.



# What the next hour computes

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## Section 1

# Dividing a computation, 1793

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# Adam Smith, 1776

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The pin maker's workshop. *Epinglier*, Diderot and d'Alembert's

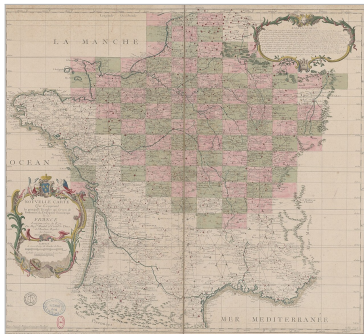
*Encyclopedie*, 1762

*One man draws out the wire, another straightens it, a third cuts it, a fourth points it, a fifth grinds it at the top for receiving the head.*

- *The Wealth of Nations*, Book I, chapter 1: **Of the Division of Labour**
- One workman making a whole pin: perhaps twenty in a day, perhaps not one
- Ten men, split across eighteen operations: **forty eight thousand**
- Same people, same tools. Only the organisation changed

# The task: the area of every parcel in France

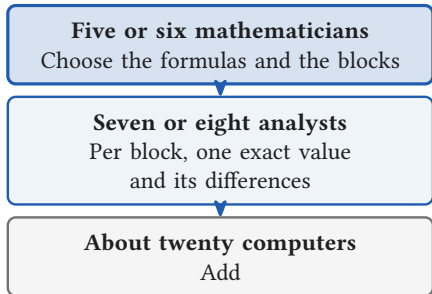
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The principal triangles of France. Maraldi and Cassini de Thury, 1744

- The Revolution replaces tax privilege with one tax on land
- So every parcel has to be measured, and its area computed
- Gaspard de Prony directs the Bureau du Cadastre, which does the computing
- Land is measured in triangles, because angles are quick and chains are slow

# Three tiers, and what each hands down



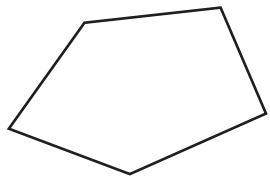
A computing room, Harvard College Observatory, about 1890

*Je conçus tout à coup l'idée d'appliquer la même méthode à l'immense travail dont je m'étais laissé imposer le fardeau, et de fabriquer mes logarithmes comme on fabrique des épingles.*

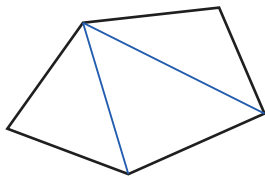
Manufacturing my logarithms as one manufactures pins.  
Gaspard de Prony, Notices, 1824, p. 5.

# One parcel, many triangles, one each

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One parcel



Cut into triangles

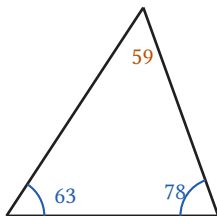


One each

- A clerk never sees a parcel. He is handed one triangle and returns one area
- Which is why twenty of them can work at once without talking to each other



# One side, two angles, and a table of logarithms



400 m

Measured with the chain

The third angle is free:

$200 - 63 - 78$

$$\log \text{Area} = 2 \log c + \log \sin A + \log \sin B \\ - \log \sin C - \log 2$$

|                    |                       |              |
|--------------------|-----------------------|--------------|
| $2 \log 400$       | +5.2041200            |              |
| + $\log \sin 63$   | -0.0778938            |              |
| + $\log \sin 78$   | -0.0264654            |              |
| - $\log \sin 59$   | -0.0970812            |              |
| - $\log 2$         | +0.3010300            |              |
| <hr/>              |                       |              |
| $\log \text{Area}$ | 4.8958120             |              |
| Area               | 78 670 m <sup>2</sup> | about 7.9 ha |

Five lookups in a seven figure table, then four additions. The parcel is ours; the units and the method are the period's.



## Section 2

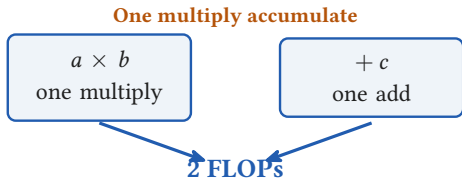
# One operation, and who does it

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2

☰ One operation, and who does it

# A FLOP is one operation on two numbers



Count, or rate

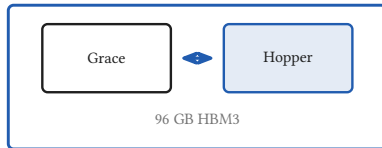
**FLOP:** a count of operations  
how much work

**FLOP/s:** operations per second  
how fast

# One computer, then and now



A computer room, NACA, 1949



One GH200. Alps holds 10,752

## Human computer, 1949

|        |                         |
|--------|-------------------------|
| Rate   | 1,000 additions a day   |
| Format | Decimal, by hand        |
| Power  | About 100 W             |
| Food   | <b>2,065 kcal a day</b> |
|        | what one person eats    |

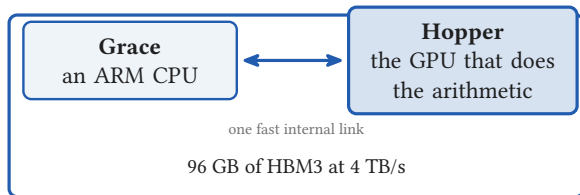
## One GH200, 2024

|                              |
|------------------------------|
| 990 TFLOP/s dense BF16       |
| 16 bits, fixed               |
| 660 W on Alps                |
| <b>13,600 kcal a day</b>     |
| what six or seven people eat |

- The clerk's rate is what the shop *delivered*. The chip's is what it *could* do

# A GH200 is a CPU and a GPU on one module

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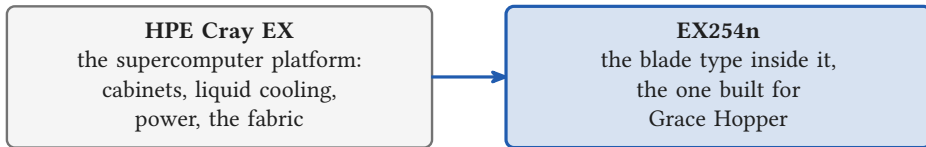


- The GPU computes. The CPU reads the data, runs Python, and keeps it fed
- HBM3 is memory stacked on the GPU package: only 96 GB, but 4 TB/s
- Between them 900 GB/s, about seven times what a plug-in card gets

# Alps is an HPE Cray EX254n

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The TOP500 entry reads: *Alps, HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS*



The trailing n marks the NVIDIA accelerated blades

- The blade choice is what makes Alps a machine for AI rather than a conventional CPU cluster

# The atelier against Alps

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| A collection of computers              | Rate              | Power      |
|----------------------------------------|-------------------|------------|
| Prony's atelier, 1795<br>twenty people | 0.694 additions/s | about 2 kW |
| Alps, 2024<br>10,752 modules           | 10.64 EFLOP/s     | 7.1 MW     |

**Fifteen billion billion times the rate, on three and a half thousand times the power**

# What the TOP500 actually measures

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A list of the 500 most powerful known supercomputers, published every June and November since 1993. Alps is rank 10 in June 2026.

It ranks them on one benchmark: **HPL**, solving a large dense linear system in **64 bit** arithmetic

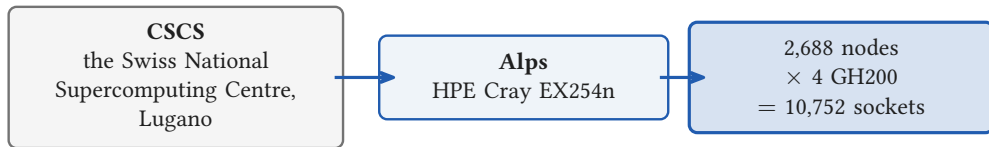
**Rmax** 434.90 PFLOP/s  
what it actually achieved

**Rpeak** 574.84 PFLOP/s  
the theoretical maximum

So even on its own benchmark it reaches 75.7 percent, and drew 7.1 MW doing it



# CSCS and Alps, the machine behind the anchor



At dense BF16 the whole partition is about **10.6 EFLOP/s**, which is the rate that matters for training.

Its TOP500 entry, 434.90 PFLOP/s at 7,124 kW, is an **fp64** measurement and is about 24 times smaller.

Machine: [top500.org/system/180259](https://top500.org/system/180259) and [cscs.ch/computers/alps](https://cscs.ch/computers/alps).

# One second of a GH200, in three registers

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## The same one second of arithmetic

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The mean age of all 8.2 billion people alive,  
one addition each

**8.3 microseconds**

130,000 photographs classified by ResNet-50,  
each 224 by 224 pixels, or 0.05 megapixels

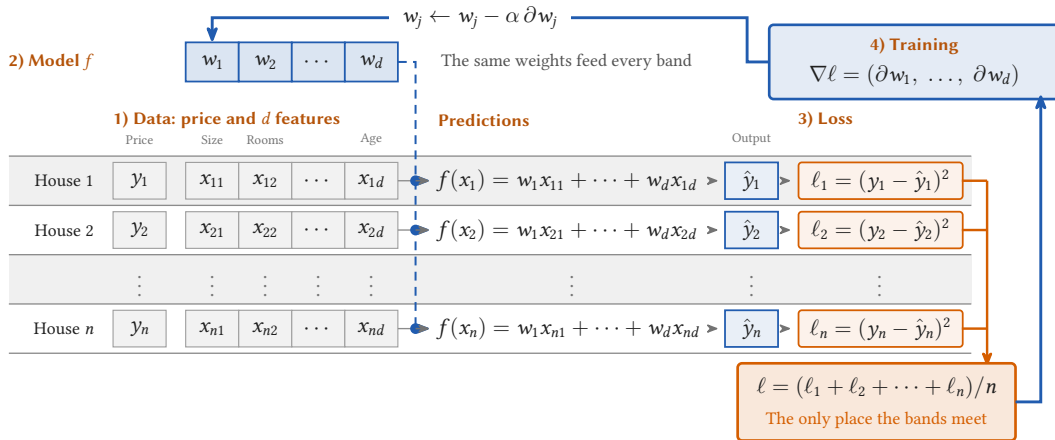
**one second**

Prony's atelier doing that same  
one second of work

**45 to 136 million years**

ResNet-50: He et al., [arXiv:1512.03385](https://arxiv.org/abs/1512.03385), 3.8 billion multiply-adds, so 7.6 GFLOP per image. A peak count.

# One training step of linear regression





### Section 3

# What the headline rate hides

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3

# One floating point number, taken apart

Start with

$$\pi = 3.14159265 \dots$$

A mantissa times a power of two

$$1.5707963 \dots \times 2^1$$

So really two numbers

mantissa 1.5707963 ...  
exponent 1

The mantissa in binary

$$1.\underline{1001001}00001 \dots$$

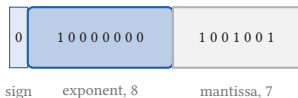
bf16 keeps seven of those bits

$$1.1001001 \times 2^1$$

the rest is thrown away

# What the seven bits cost

The sixteen bits



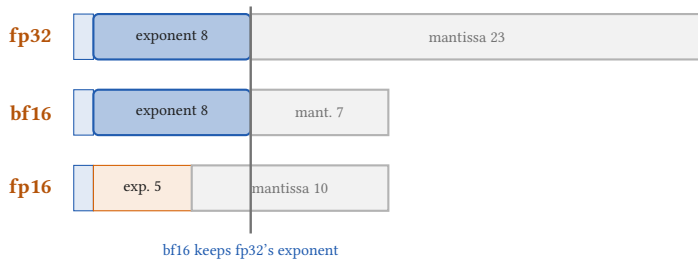
Back to decimal  
3.140625

We started from  
3.14159265...

Lost 0.00097, or 0.03 percent

- Seven bits cannot hold  $\pi$ , so the number stored is not the number you asked for

# Not every FLOP costs the same



# Why we say dense BF16, and not just BF16

The datasheet's fast number assumes a pattern



Two of every four pinned to zero, so the hardware skips them  
1,979 TFLOP/s

Pretraining cannot promise it



Every weight starts random and has to stay free to move  
990 TFLOP/s per GH200, dense

- Sparsity is just something you impose on a finished model



# Capacity and bandwidth are different numbers

**96 GB**

how much fits

Decides whether the weights and the optimiser state fit on one device at all

**4 TB/s**

how fast you can move it

Decides how quickly the arithmetic units can be fed once the data is there

- 96 GB is less than an ordinary server holds. That is the trade: HBM is stacked on the package and wired short and wide, which is what buys the speed

# Two hundred and forty six operations per byte



$$\frac{990 \times 10^{12} \text{ FLOP/s}}{4.023 \times 10^{12} \text{ bytes/s}} = 246 \text{ FLOP per byte}$$

A rate over a rate: the seconds cancel, and what is left is operations per byte

So unless the computation reuses each byte about that many times, the units wait. A large matrix multiply does reuse. Many other operations do not.

- This is one of the reasons a real run reaches 21 to 46 percent of peak



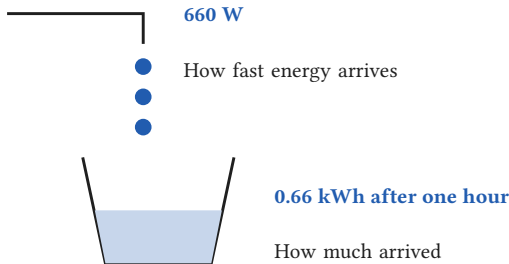
#### Section 4

# The machine, and what it draws

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4

# A watt is a rate, a kilowatt hour is an amount



## Watch the slash

km/h the slash divides,  
so it is a speed

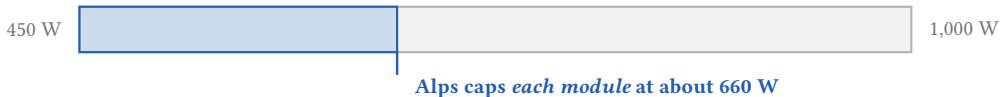
kWh no slash. The hour  
**multiplies**, so it is an amount

Longer gives you *more*, so the hour  
cannot be dividing

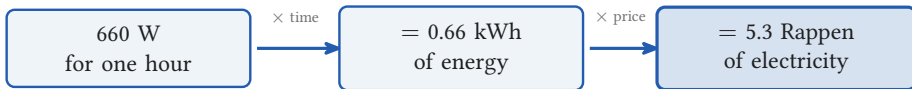
- Rate  $\times$  time = amount, twice over: FLOP/s  $\times$  seconds = FLOPs, and watts  $\times$  hours = kWh

# A wattage becomes a bill in two multiplications

One module is just a band



The operator chooses: near 1,000 W is faster, lower saves energy and cooling

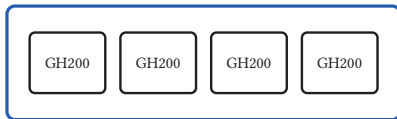


450 to 1,000 W: GH200 datasheet. 660 W: the Alps cap, [docs.cscs.ch/running/slurm](https://docs.cscs.ch/running/slurm). 8 Rappen per kWh assumed (ElCom 2026 large consumer band 6.75 to 9; households pay 27.7).  
Check:  $10,752 \times 660 \text{ W}$  is 7,096 kW against the 7,124 kW TOP500 measured during HPL, [top500.org/system/180259](https://top500.org/system/180259).

- That is **one module**. All 10,752 is 7.1 megawatts, about 570 francs an hour
- The whole Apertus run is 3.96 GWh, about 0.3 million francs

# Nodes and sockets

One node



**Socket** one processor module  
**Node** one independent computer

One operating system, one address space

$2,688 \text{ nodes} \times 4 \text{ modules} = 10,752 \text{ sockets}$

Socket to socket, inside the node: 150 GB/s

Between nodes: 25 GB/s per GPU, six times less

- The node boundary is where the bandwidth falls off

# Four links, and each step down costs you

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- **4,023 GB/s**    the GPU to its own HBM3
- **900 GB/s**    Grace to Hopper, inside one module
- **150 GB/s**    socket to socket, inside a node
- **25 GB/s per GPU**    between nodes

**Each step down costs about a factor of five. Top to bottom, 161 times.**

- So you keep the data at the fast end and cross the slow links as rarely as you can



## Section 5

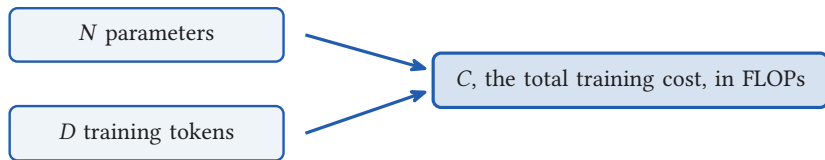
# From a model to GPU hours

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5



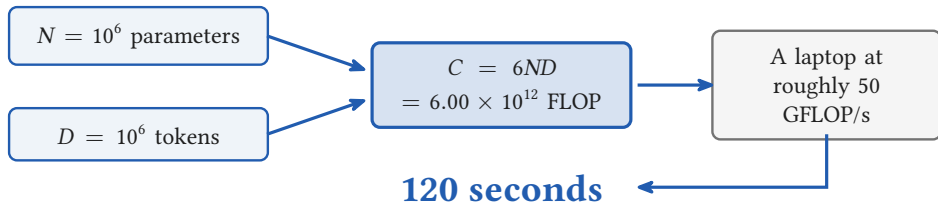
# Two numbers price a training run



$$C = \underbrace{6}_{\text{per parameter, per token}} \times \underbrace{N}_{\text{parameters}} \times \underbrace{D}_{\text{tokens}}$$

[Kaplan et al. 2020](#), section 2.1. Where the 6 comes from is the 12:00 lecture; for the next fifty minutes it is a rule of thumb.

# A million by a million is two minutes



# The anchor: Apertus, trained on Alps

## Apertus-70B

Released 2 September 2025, Apache 2.0  
 $70.60 \times 10^9$  parameters, 15 trillion tokens  
1,811 languages, 40 percent of it not English

## Alps, at CSCS Lugano

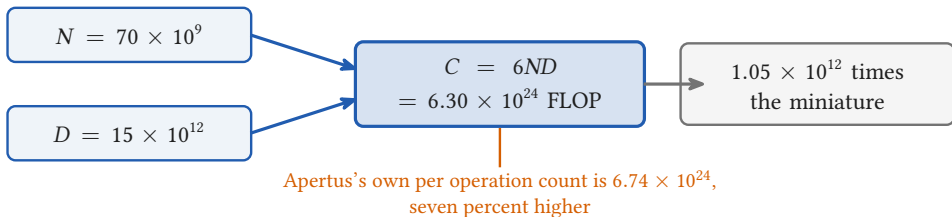
10,752 GH200 sockets, 4 per node  
990 TFLOP/s dense BF16 each  
TOP500: 434.90 PFlop/s at 7,124 kW

**The production run** Over 6 million GPU hours, over roughly three months, on up to 4,096 GPUs, sustaining about 723 tokens per second per GPU

Built by EPFL, ETH Zurich and CSCS. Every number in the next forty minutes is about this run.

Model: [arXiv:2509.14233](https://arxiv.org/abs/2509.14233). Run: [arXiv:2604.12973](https://arxiv.org/abs/2604.12973). Machine: [top500.org/system/180259](https://top500.org/system/180259).

## The same formula on Apertus-70B



# From one GPU hour to the whole run

The run: training Apertus-70B on 15 trillion tokens, on up to 4,096 GPUs.

4,096 GPUs flat out  
for **61 days**

It spanned about 91 days,  
so **2,747** on average

The gap is ramp up, restarts and smaller phases: a campaign is not one uninterrupted run at full width

6 million GPU hours  $\times$  660 W = **3.96 GWh**  $\times$  8 Rappen = **0.32 million francs**



## Section 6

# Peak, and what a real run gets

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6

# Counting the workshop is not measuring it



**Twenty computers, one desk each**

Counted from the doorstep. Nothing was run.

**Now watch for an hour**

A clerk waits for the next sheet.

Two clerks stop to compare a figure.

The batch is small, so some desks are empty.

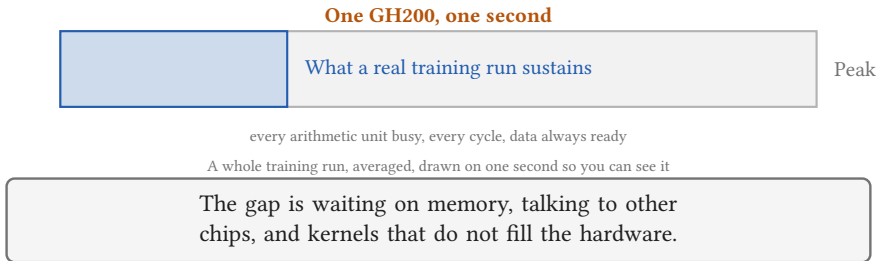
Count the desks,  
multiply by how fast  
one clerk works



That is the workshop's  
**peak**

- The clerks were at full speed the whole time. They were not always given something to do

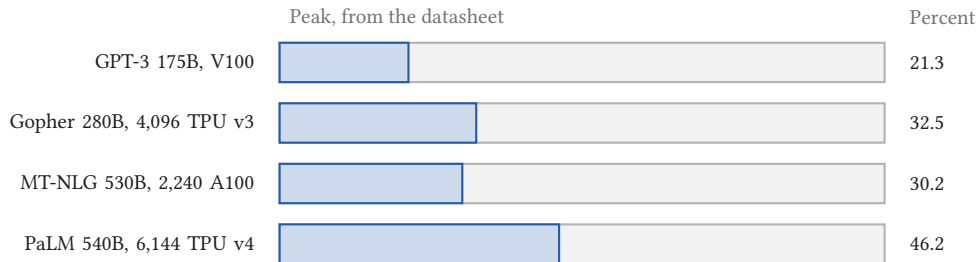
# Peak is just a ceiling



- Nobody reaches peak. The datasheet number is a ceiling
- The share you do reach is [model FLOPs utilisation](#), written MFU
- It is the blue bar over the whole bar, so MFU 0.30 means thirty percent of peak



## Published runs reach 21 to 46 percent of peak



- Two ran on TPUs, so the umbrella word is [accelerator](#): GPU, TPU, or GH200
- Model FLOPs utilisation: the share of peak a training run reaches
- Assuming 100 percent is wrong by a factor of two to five

## Where the utilisation formula comes from

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$$\text{MFU} = \frac{6ND}{\text{GPU-hours} \times 3,600 \text{ s/h} \times \text{peak FLOP/s}}$$

- Tokens per second is what you observe. FLOP per token is what you count
- Substitute  $D/T$  tokens per second, and  $6N$  FLOP per token, where  $T$  is the wall clock duration of the training run
- $T$  times chips is exactly GPU-seconds, which is what the run reports
- So the measurement enters through the hours, and  $6ND$  supplies the work

# Six million hours, measured against peak

## The consumption

Over 6 million GPU hours, the 70B production run

measured  
× datasheet

$$6 \times 10^6 \text{ GPU-h} \times 3,600 \text{ s/h} \times 990 \text{ TFLOP/s per GPU} \\ = 2.14 \times 10^{25} \text{ FLOP of peak capacity}$$

analytic

$$\frac{6.30 \times 10^{24}}{2.14 \times 10^{25}} = 29.5 \text{ percent}$$

6ND for the 70B



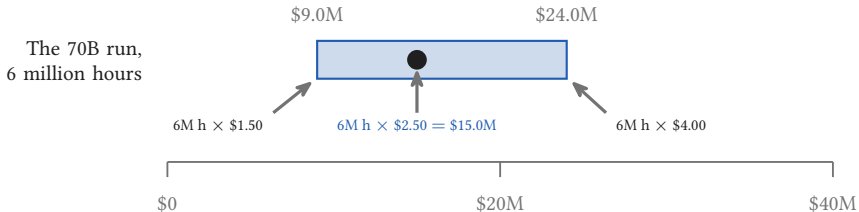
## Section 7

# Dollars and energy

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# Six million hours, priced at cloud rates



The \$1.50 to \$4.00 band is an H100 rental survey, November 2025.

# The same hours, in gigawatt hours

8.85 million GPU hours  
(90 days × 4,096 GPUs)

×

560 W per module

=

4.95 GWh

4.95

3.4



8.6

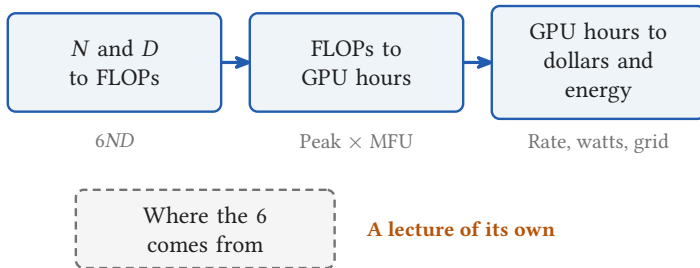


- About 1,100 Swiss households for a year, at 4,500 kWh each

Range 750 to 1,900 households, over four sourced module figures: **560 W** tech report, [arXiv:2509.14233](https://arxiv.org/abs/2509.14233); **660 W** CSCS cap, [docs.cscs.ch/running/slurm](https://docs.cscs.ch/running/slurm); **855 W** from the TOP500 run, [top500.org/system/180259](https://top500.org/system/180259); datasheet band 450 to 1,000 W.

Household 4,500 kWh a year is ElCom's H4 profile. CSCS publish a PUE below 1.2, cooling from Lake Lugano ([cscs.ch](https://cscs.ch)).

# What you can price



A lecture of its own

- You can now price a training run on the back of an envelope
- In the exercise you build the envelope